RetailPulse

AI-Powered Customer Analytics & Demand Forecasting Platform

FINAL PROJECT REPORT

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PROJECT OVERVIEW

A retail analytics prototype covering transaction data preparation, exploratory analysis, customer segmentation, sales forecasting, churn-risk modelling, inventory planning, data-quality monitoring, and a multi-page Streamlit dashboard.

Dataset: Online Retail transaction history, 2009–2011

Report date: 19 October 2026

1. Executive Summary

RetailPulse is a retail analytics prototype that turns historical transaction records into decision-support views. Implemented work includes data preparation and validation, EDA, RFM segmentation using K-Means and DBSCAN, Prophet and LSTM forecasts, ensemble comparison, XGBoost churn prediction with SHAP, historical-rate inventory recommendations, monitoring diagnostics, and a multi-page Streamlit dashboard.

Area	Evidence	Status
Data	1,067,371 rows; cleaned 24-column output; 36 quality checks passed	Implemented
Segmentation	5,878 RFM-eligible customers; K-Means k=6; PCA and DBSCAN	Implemented
Forecasting	Prophet, LSTM and equal-weight ensemble on 30-day holdout	Prophet best; target unmet
Churn	XGBoost, temporal validation, Optuna and SHAP	ROC-AUC target unmet
Inventory	4,916 products reviewed; 378 confidence flags	Planning estimate only
Monitoring	Sales distribution and churn-score diagnostics	Caveats documented

Overall: a functioning analytical prototype, not a validated production system. Several acceptance targets and deployment tasks remain outstanding.

2. Problem, Objectives and Acceptance Targets

Retail teams need to understand product demand, customer value and inactivity, and likely near-term sales. Transaction records alone do not provide stock-on-hand, unit costs, supplier lead times, or complete customer identity.

- Build repeatable ingestion, cleaning, feature engineering and data validation.
- Describe sales trends, distributions, missing values and correlations.
- Segment customers using RFM features and compare K-Means with DBSCAN.
- Compare Prophet and LSTM on a chronological holdout.
- Estimate churn risk and explain influential features.
- Produce inventory review signals with uncertainty flags.
- Present results and downloadable artifacts through Streamlit.

Targets included forecast MAPE ≤12%, churn ROC-AUC ≥0.88, precision@top 20% ≥0.75, and inventory impact targets. These are goals, not achieved outcomes unless supported by evidence.

3. Dataset and Data Preparation

The source workbook contains 525,461 rows in the 2009–2010 sheet and 541,910 in the 2010–2011 sheet, totalling 1,067,371. Processed dates range from 1 December 2009 through 9 December 2011.

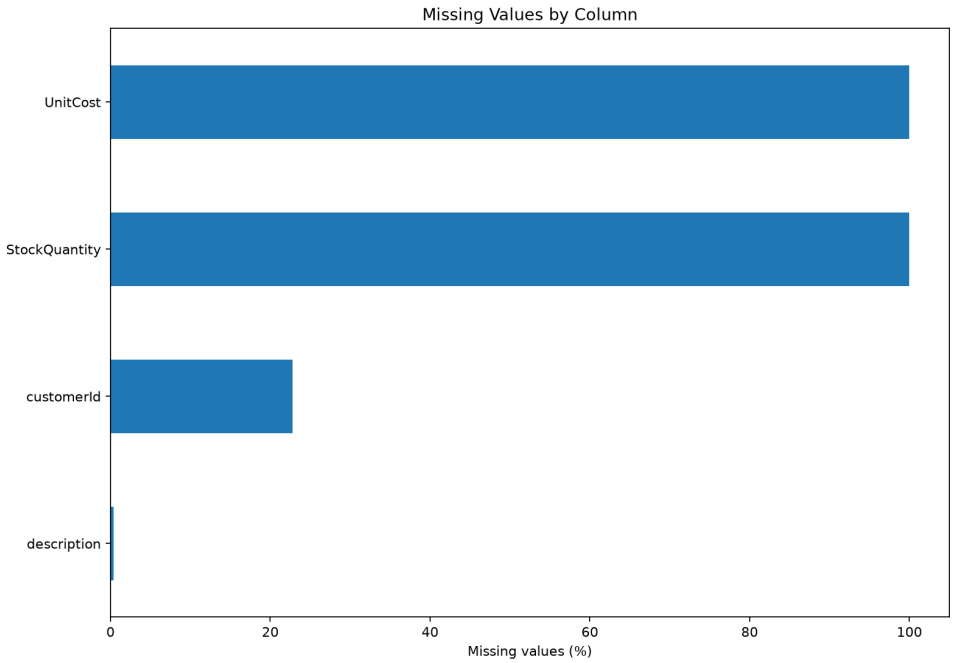


Figure. Missing-value profile.

Audit item	Observed result
Cleaned data	1,067,371 rows × 24 columns
Unique product identifiers	5,304
Identified customers	5,942 in EDA; 5,878 RFM-eligible
Missing customer IDs	243,007 rows (~22.77%)
Missing descriptions	4,382 rows (~0.41%)
Invalid dates	0 flagged
Non-positive quantity / price	22,950 / 6,207 rows
Cancellation invoice rows	19,494
Duplicate candidates	12,133 within-sheet; 22,523 cross-sheet overlap rows

The sheets overlap on 1–9 December 2010. Overlapping records were flagged rather than indiscriminately deleted. A churn sensitivity comparison found only small changes in validation metrics. UnitCost and StockQuantity are unavailable.

4. Exploratory Data Analysis

EDA reviewed temporal sales patterns, geographic concentration, transaction distributions, product categories, missingness, and numeric correlations. These are descriptive diagnostics, not evidence of causal relationships.

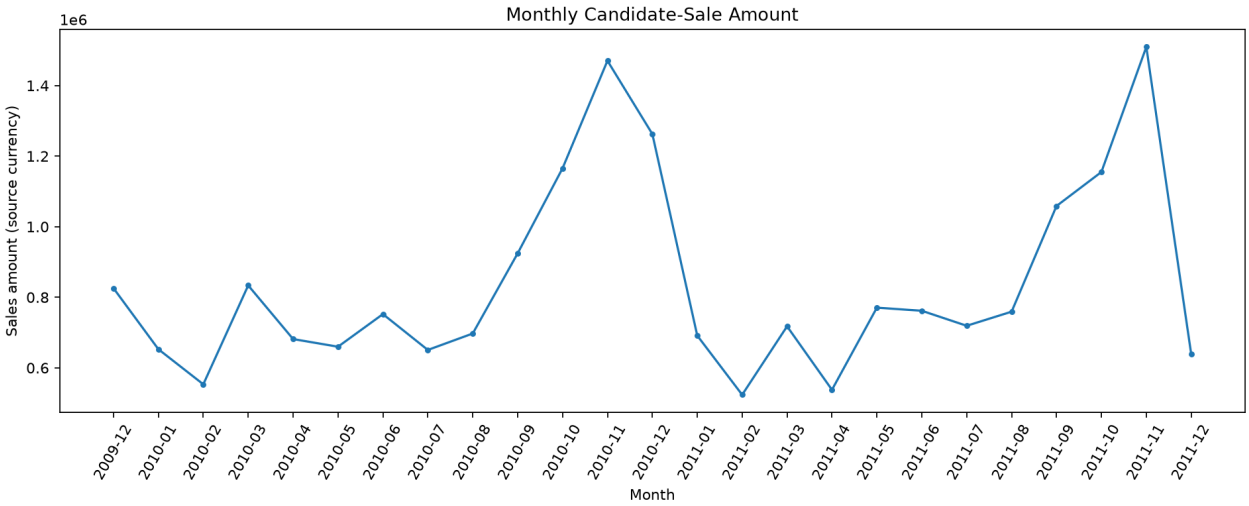


Figure. Monthly sales over the observed period.

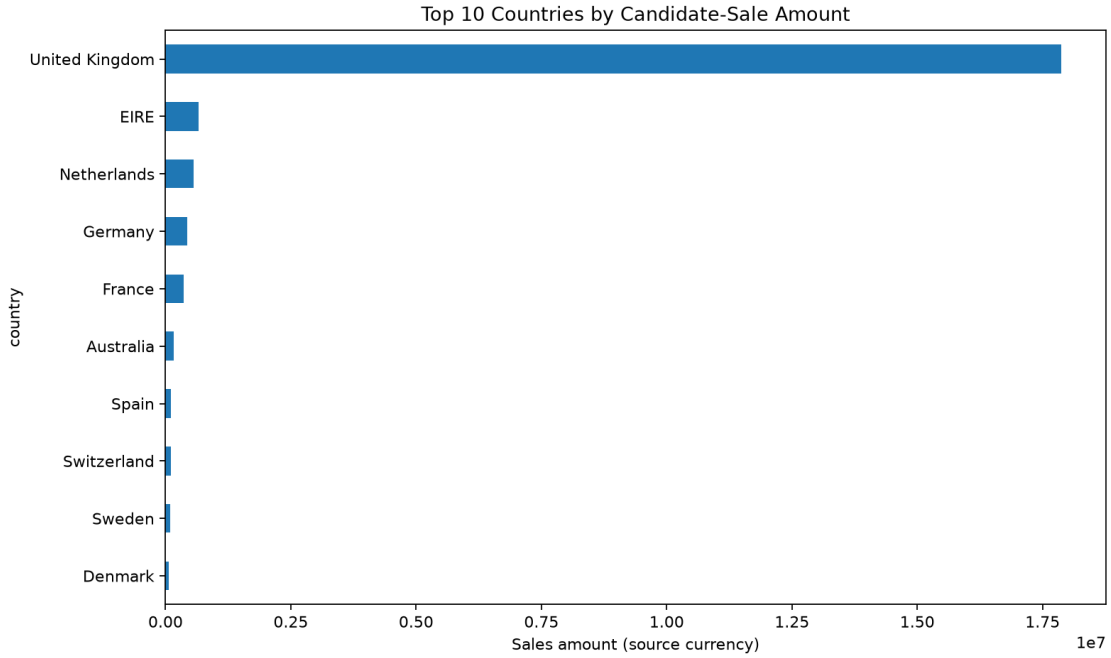


Figure. Countries with highest recorded sales.

4. Exploratory Data Analysis (continued)

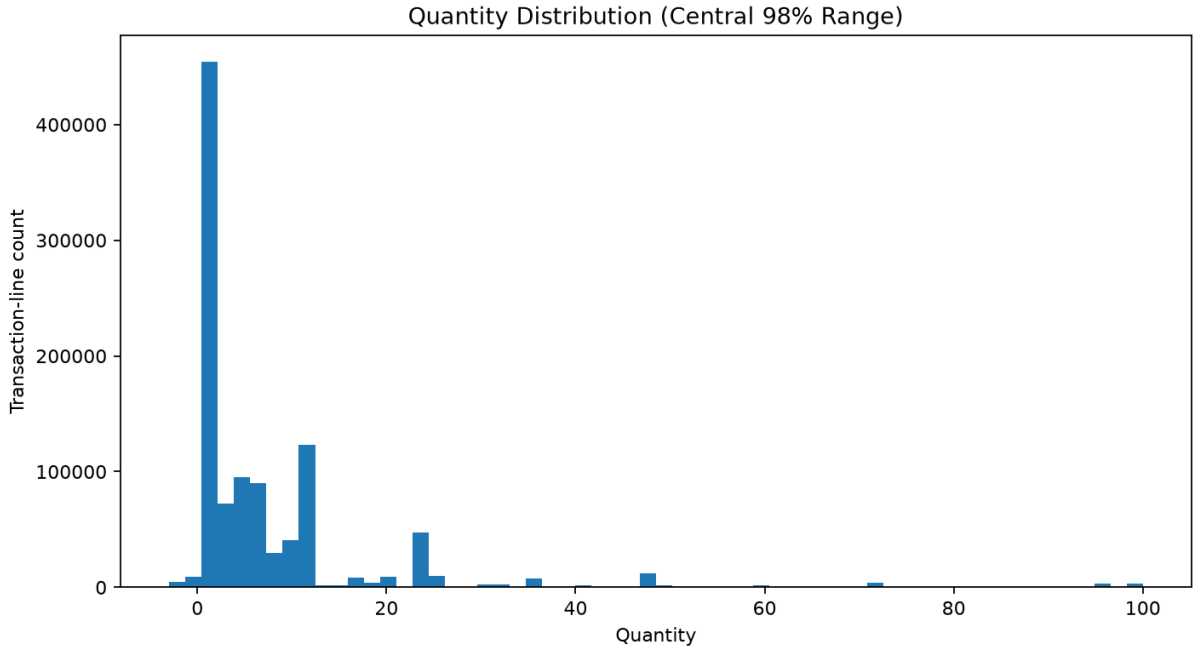


Figure. Transaction quantity distribution.

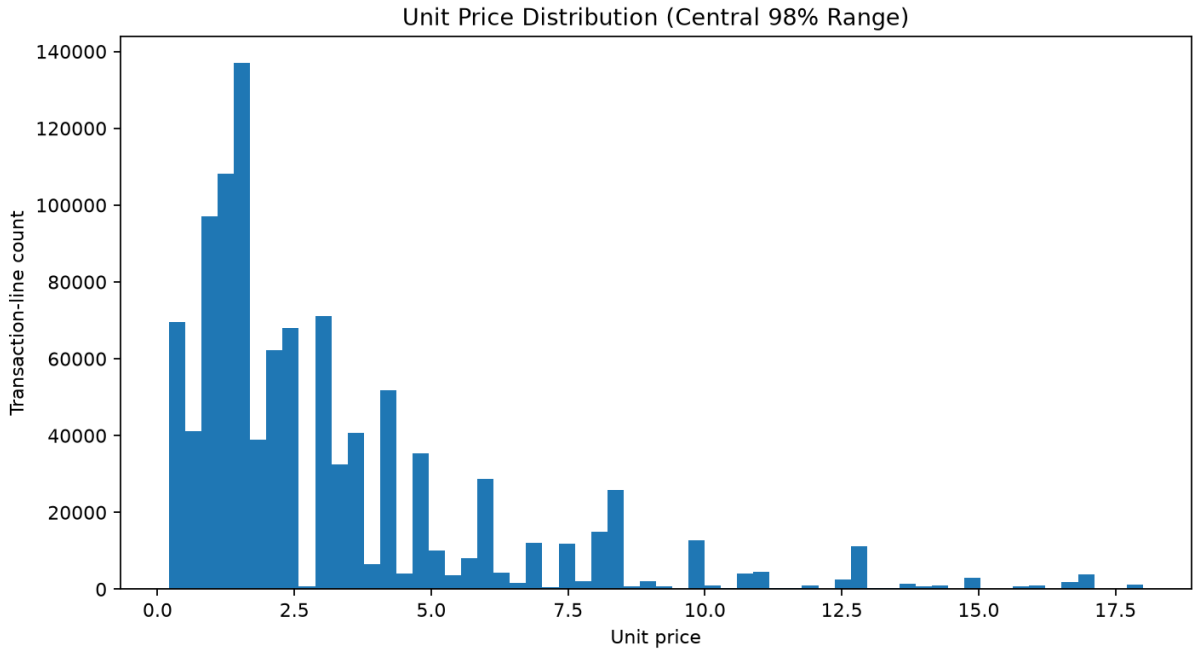


Figure. Unit-price distribution.

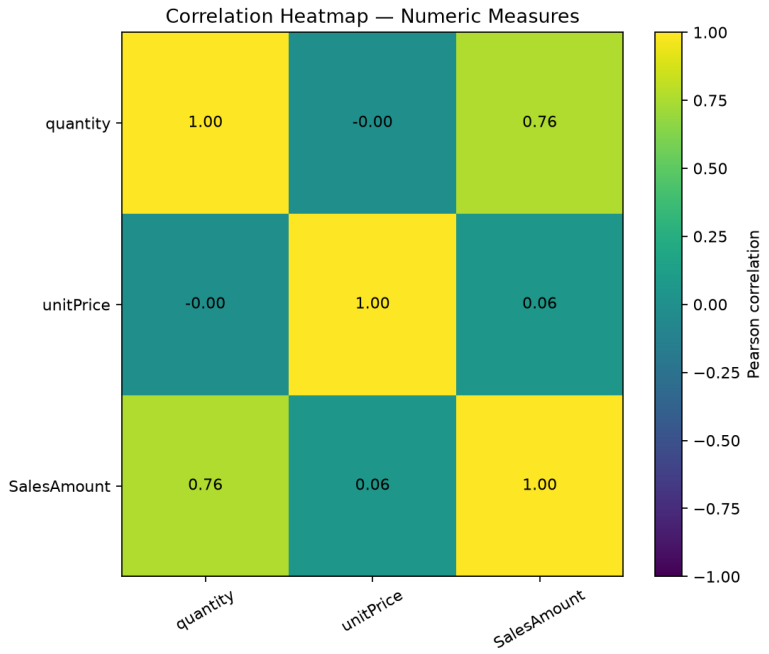


Figure. Correlation heatmap of numeric variables.

The current categorisation logic leaves 566,838 product transaction rows uncategorised. Recorded SalesAmount totals approximately 19,287,250.57 in the dataset currency context; this is not independently audited revenue.

5. Customer Segmentation

RFM features were calculated for eligible customers. Log1p transforms and StandardScaler were applied. K-Means candidates k=2–8 were evaluated. Although k=2 had the highest silhouette score (~0.438), k=6 (~0.333) was selected for richer business interpretation.

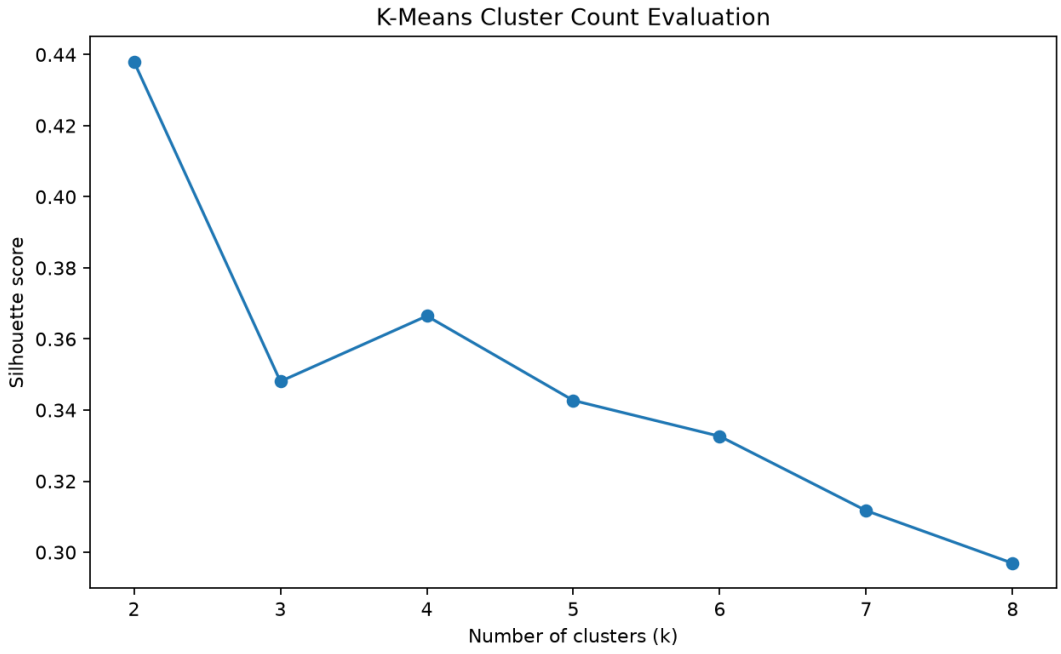


Figure. Silhouette score across candidate cluster counts.

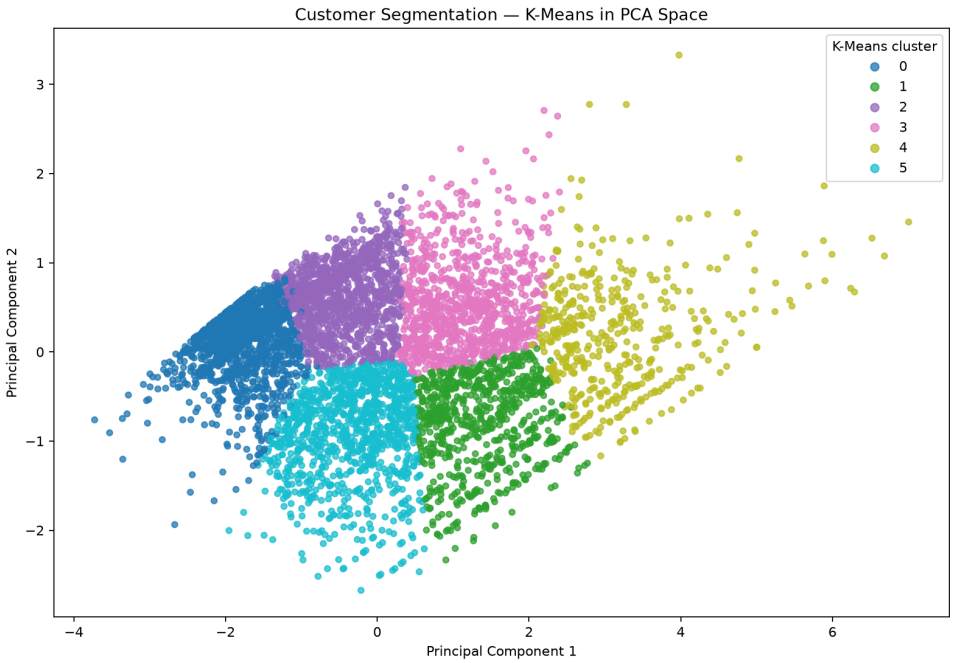


Figure. K-Means segments projected onto PCA components.

5. Customer Segmentation (continued)

The first two PCA components explain 95.136% of variance in the transformed feature space. PCA supports visualisation; its axes are not original RFM variables.



Figure. DBSCAN groups and noise points in PCA space.

Cluster	Customers	Median recency	Frequency	Monetary	Interpretation
0	1,471	426 days	1	223.98	Lapsed, low-value
1	712	11 days	6	1,940.38	Recent and active
2	1,251	313 days	3	960.63	At-risk / cooling
3	938	67 days	8	3,056.86	Established valuable
4	458	9 days	23	9,921.98	High-value loyal

Cluster	Customers	Median recency	Frequency	Monetary	Interpretation
5	1,048	37 days	2	500.77	Recent, low-frequency

Cluster 4 represents ~7.8% of customers and ~50.3% of segment monetary total. DBSCAN marked 94 customers (~1.6%) as noise; unusual points should be reviewed, not automatically discarded.

6. Forecasting: Preparation and Methods

Daily sales were aggregated into 739 calendar days. Missing dates were filled with zero. The Augmented Dickey–Fuller p-value was ~0.2836, so the analysis did not reject the unit-root null at common significance levels. Weekly decomposition and autocorrelation plots were generated.

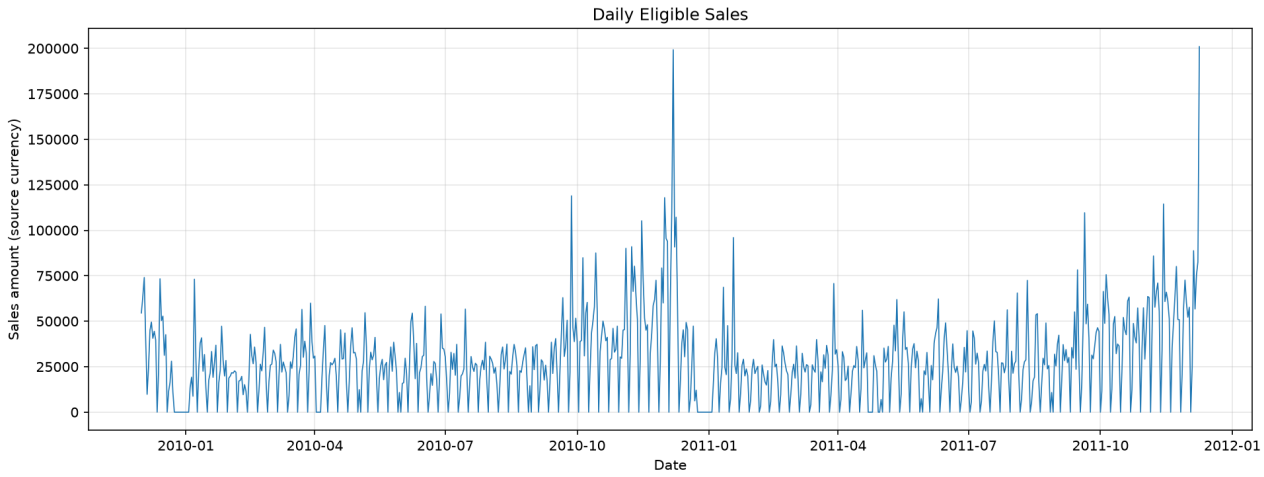


Figure. Daily sales series.

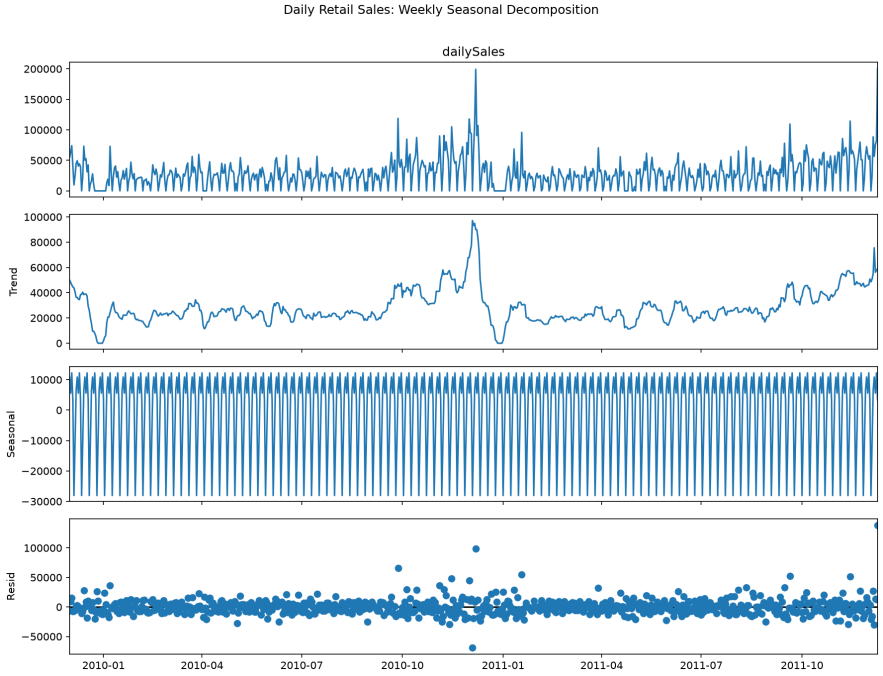


Figure. Additive decomposition using a seven-day period.

6. Forecasting: Baseline Results

The final 30 days (10 November–9 December 2011) were held out; 709 earlier observations were used for training. MAPE is calculated on non-zero actual-sales days because zero actuals make percentage error undefined or unstable.

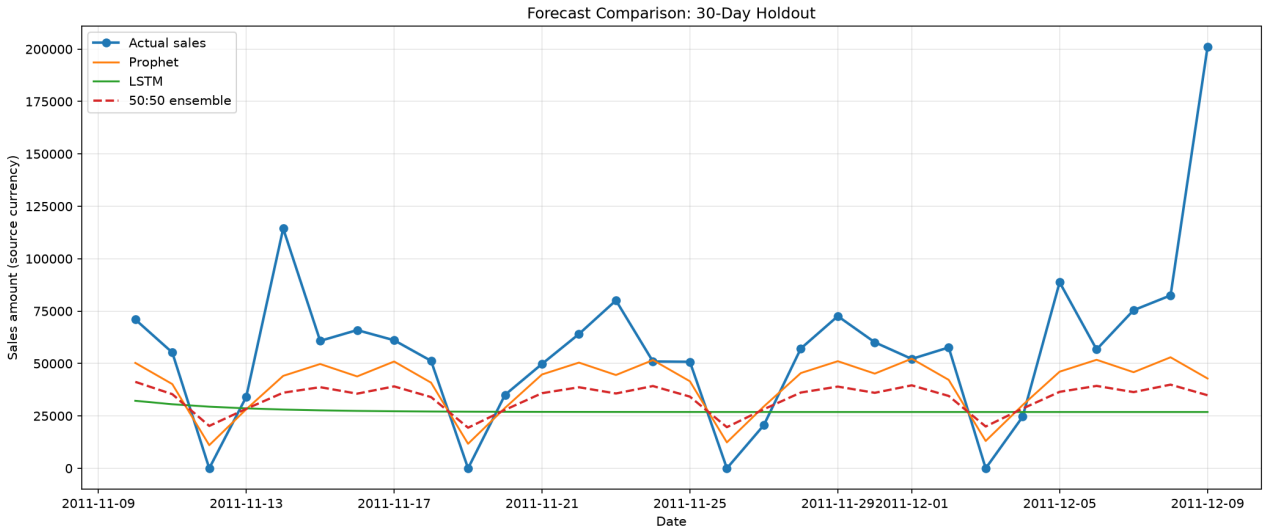


Figure. Model error comparison.

Model	MAE	RMSE	MAPE (non-zero)	WAPE
Prophet	20,906.80	35,770.99	27.50%	37.02%
LSTM	36,949.82	48,005.31	51.83%	65.43%
50:50 ensemble	28,902.86	41,373.56	39.61%	51.18%

Prophet is the strongest current baseline. The equal-weight ensemble is worse than Prophet alone. The MAPE target of 12% has not been met.

6. Forecasting: Model Diagnostics

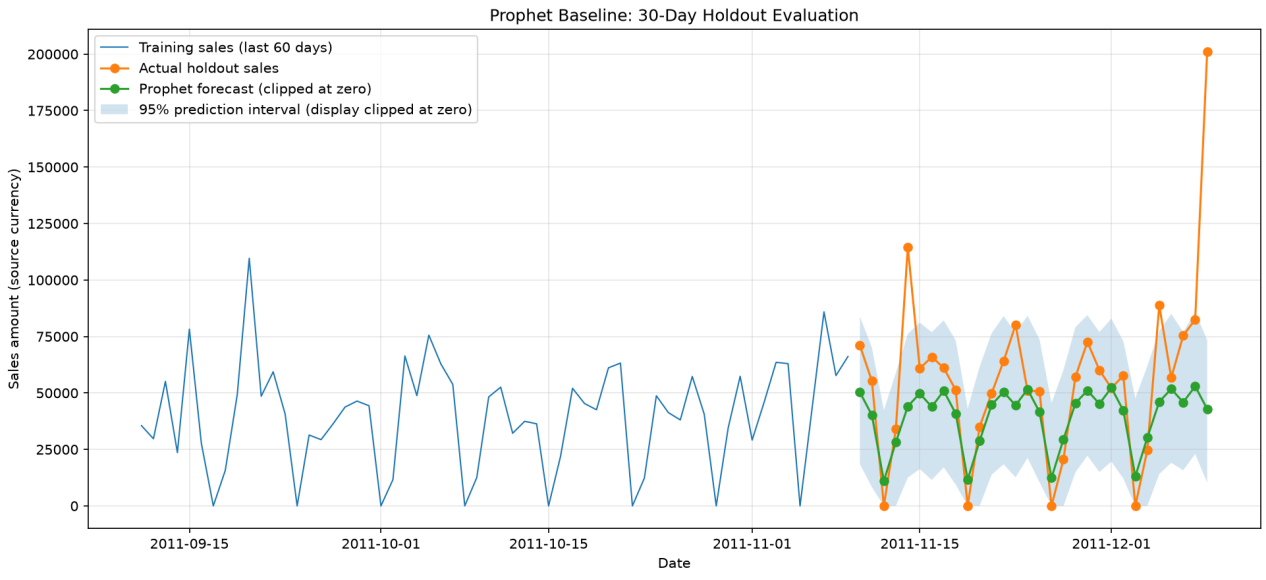


Figure. Prophet forecast versus actual sales.

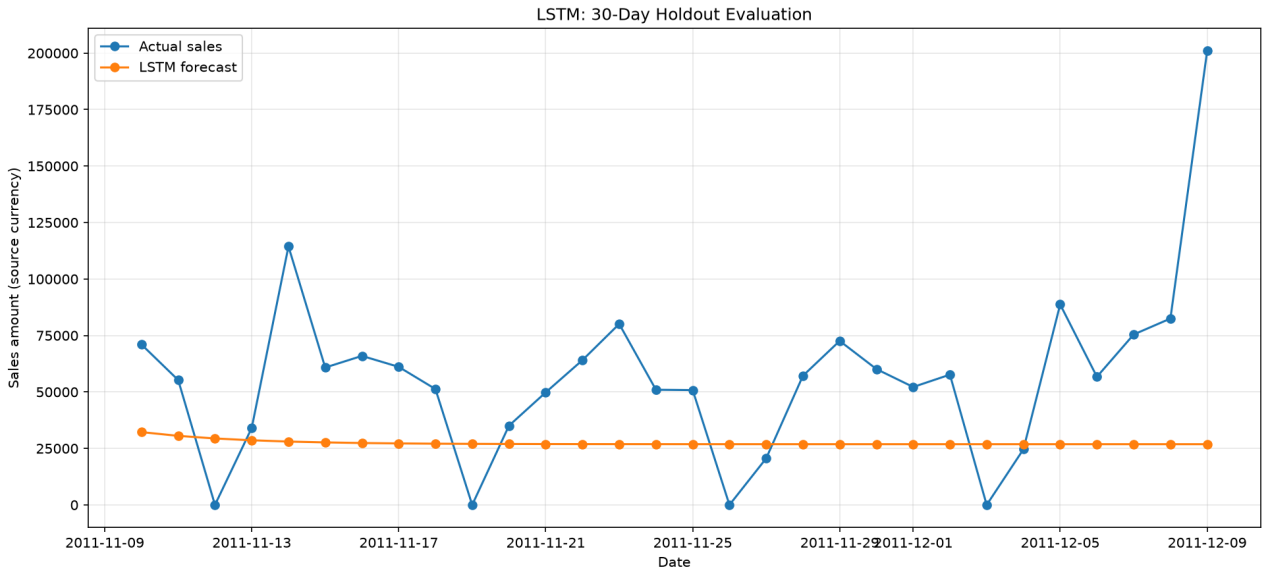


Figure. Recursive LSTM predictions versus actual sales.

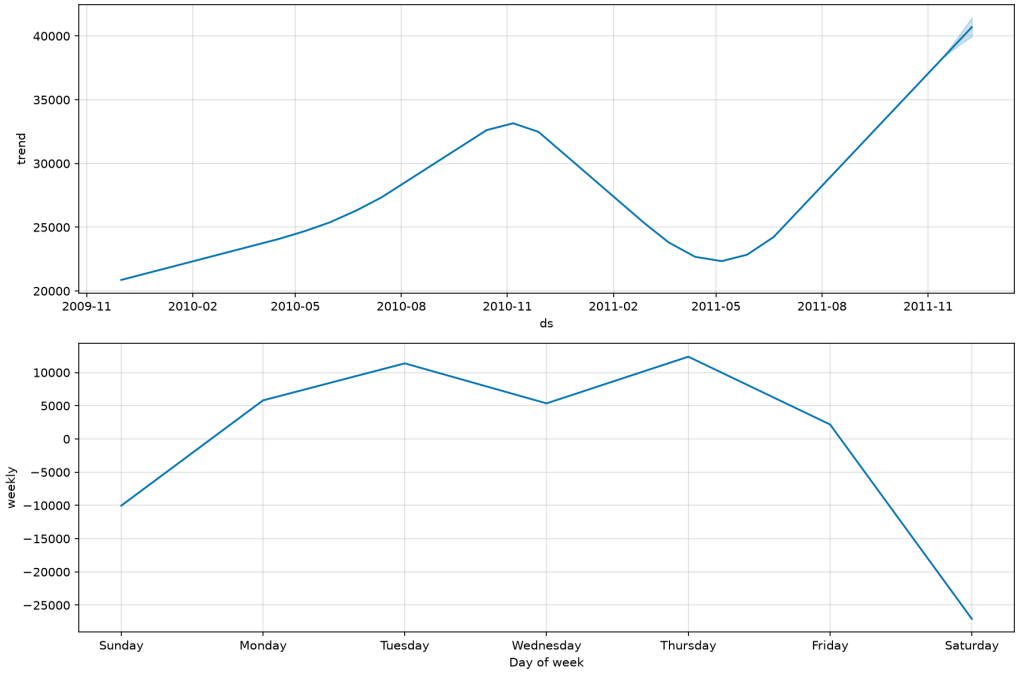


Figure. Prophet trend and seasonal components.

The LSTM used a 30-day input sequence, 32-unit hidden layer, training-only scaling, and a validation window from training history. Short history and volatile totals limit reliability. Forecasts are planning signals, not precise commitments.

7. Churn-Risk Prediction

Churn was defined as no qualifying purchase in the following 90-day label window, using a 365-day feature window. A random-split baseline and separate temporal validation were evaluated. Three temporal cutoffs were used; the latest was the test period.

Evaluation	ROC-AUC	Precision@top 20%	Notes
Random-split baseline	0.7682	0.7731	Exploratory split
Temporal baseline	0.7633	0.7381	One temporal test cutoff
Optuna temporal model	0.7695	0.7555	30 trials; latest snapshot

ROC-AUC target 0.88 is not met. Precision@top 20% exceeds 0.75 for the tuned temporal result but not the temporal baseline. Results should not be assumed to generalise to future periods.

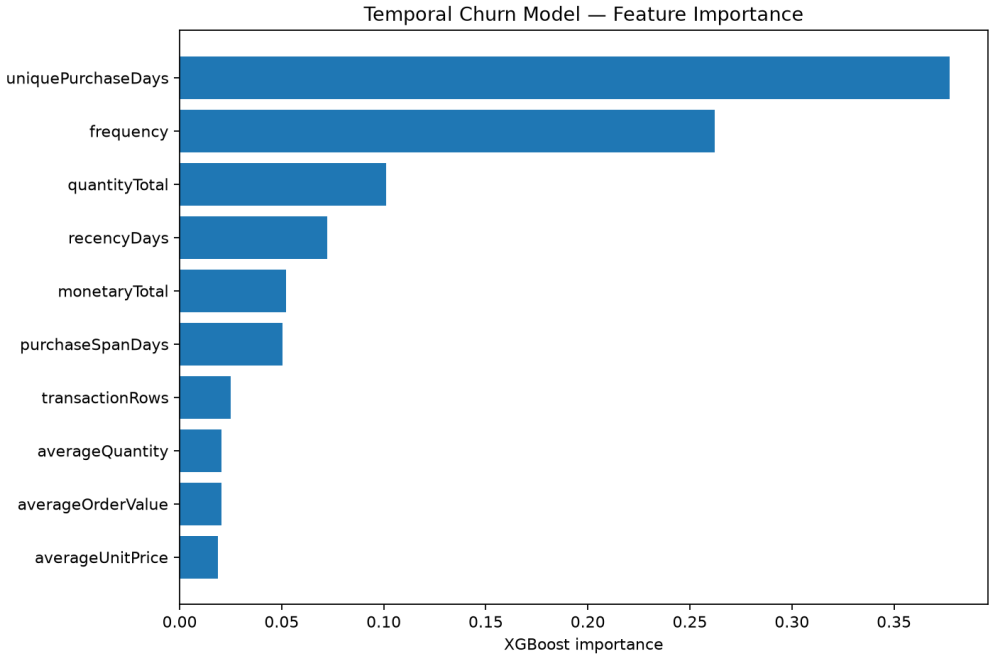


Figure. Temporal model feature importance.

7. Churn-Risk Prediction: Explainability

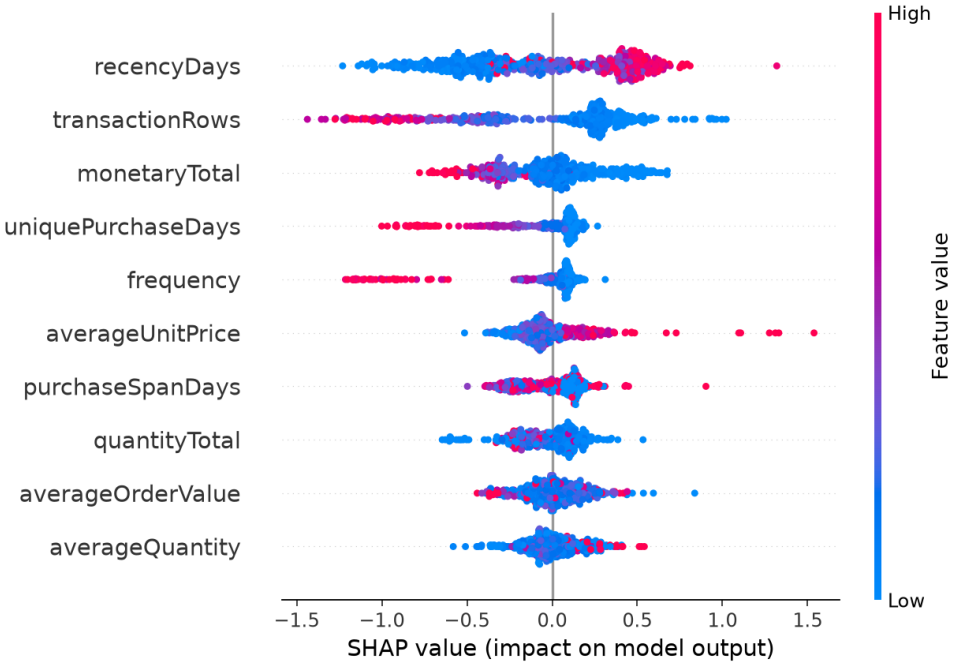


Figure. SHAP summary view.

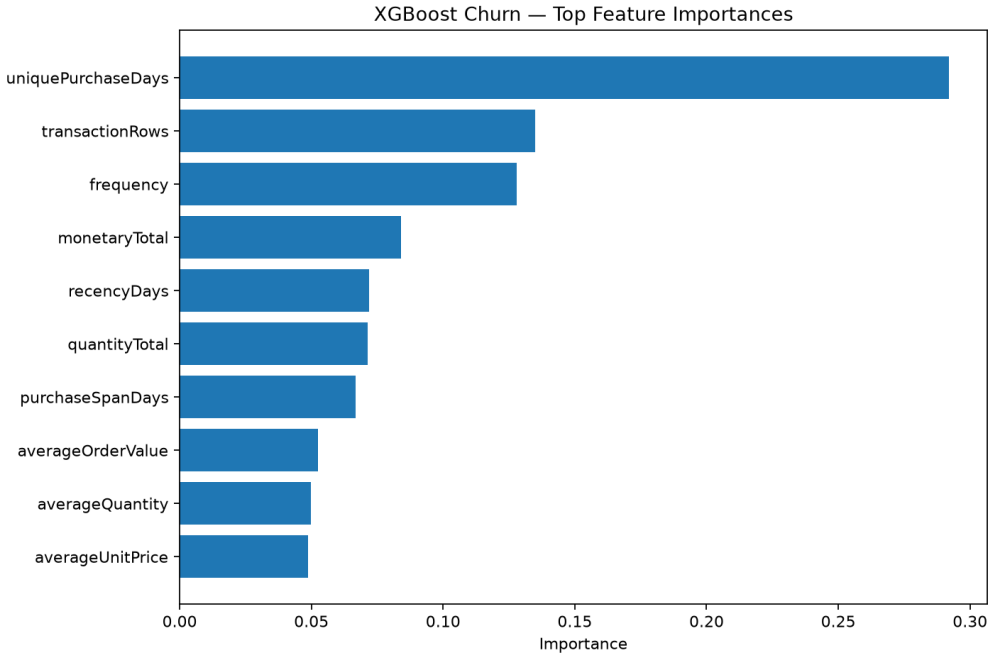


Figure. Model feature importance.

Feature importance describes model use of variables; SHAP estimates feature contributions under the selected explainer. Neither is causal. Baseline importance ranked uniquePurchaseDays, transactionRows, frequency, monetaryTotal, and recencyDays highly. SHAP highlighted recencyDays and transactionRows, followed by monetaryTotal, uniquePurchaseDays, and frequency.

8. Inventory Planning and Demand Confidence

Inventory recommendations use historical product sales rates. The workflow reviewed 4,916 products; 2,896 had sales in the most recent 28-day window. Assumptions include 14-day lead time, seven-day review interval, 30-day horizon, and approximate 95% service level.

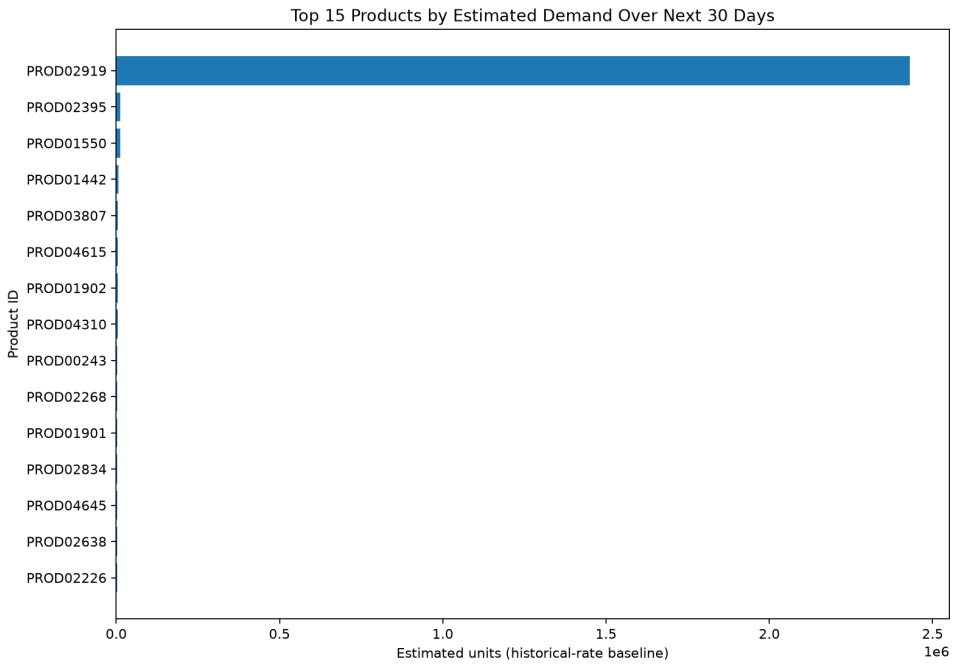


Figure. Products with highest estimated 30-day demand.

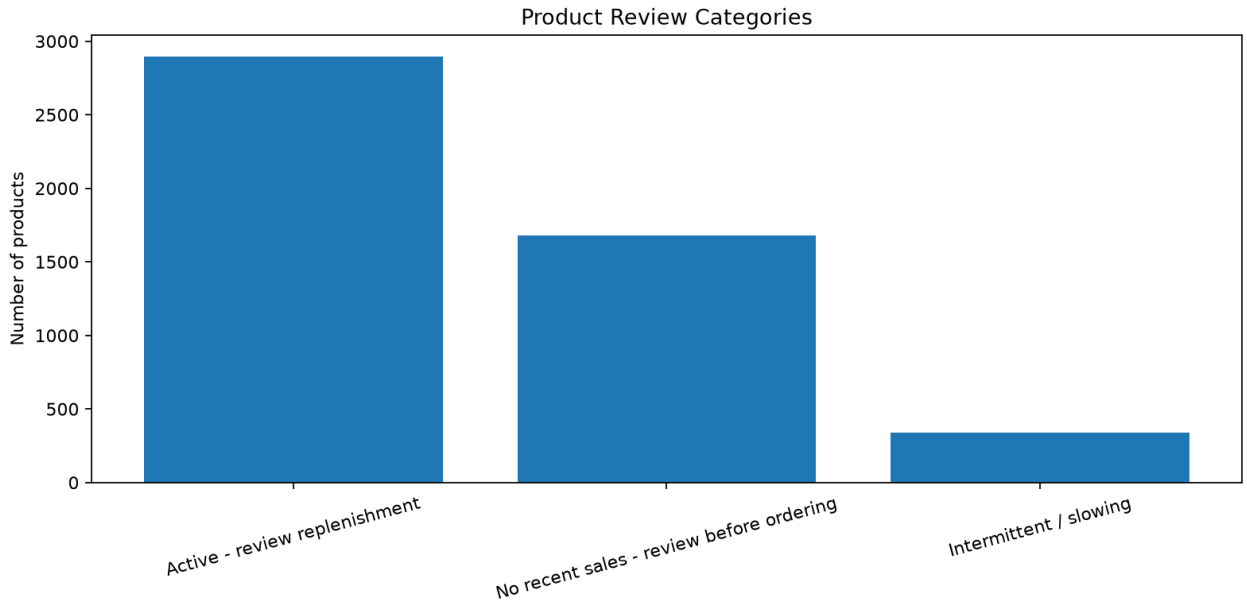


Figure. Inventory review categories.

8. Inventory Planning: Limitations

The demand-confidence audit flagged 378/4,916 products (7.69%) for manual review due to low active-day coverage or concentrated/spiking demand.

Product PROD02919 recorded 80,995 units on 9 December 2011 in one invoice row, representing the entire recent 28-day quantity. The row was not marked as an obvious duplicate or invalid record; it was retained rather than arbitrarily capped, but must be verified before operational use.

Limitation	Consequence
No stock-on-hand	Cannot compare recommendations with actual inventory position.
No unit costs	Cannot quantify holding cost, stockout cost, or margin impact.
Historical-rate method	No separately validated product-level Prophet/LSTM forecast.
Demand spikes	One bulk order can dominate recent rates.
No intervention outcomes	Stockout reduction, savings, or revenue lift are unverified.

9. Data Quality, Monitoring and Reproducibility

Great Expectations validation passed 36 configured checks with zero failures across 1,067,371 rows and 24 columns. Passing configured expectations does not prove every business-data issue has been detected.

- Sales drift compared 517 reference observations with 222 current observations; dailySales was flagged as changed (KS p≈9.20×10■■). Distribution change does not itself prove model degradation.
- The churn-score diagnostic split records by row order rather than a strictly chronological population, so it is not conclusive evidence of temporal drift.
- Monitoring completed with caveats; those caveats are retained.
- Local JSONL experiment records include parameters, metrics, artifact paths, sizes and SHA-256 hashes. Existing MLflow artifacts/database were retained as historical records.

Interpretation

Monitoring is prototype-level. Reference windows and population definitions should be explicitly chronological and reviewed before operational alerting.

10. Dashboard and User Workflow

Page	Purpose
Executive Overview	High-level business and model indicators.
Analytical Overview	Brief project introduction.
Sales & Forecasting	History, holdout comparisons and forecast metrics.
Customer Segmentation	K-Means segments, PCA, DBSCAN and noise.
Churn Risk	Risk predictions, temporal metrics, SHAP/feature importance.
Inventory Planning	Demand estimates, review categories and confidence flags.
Data Quality & Exports	Quality diagnostics and available downloads.

The local app.py syntax check passed. Source inspection confirmed SHAP/feature-importance display, inventory review flags, and warnings about missing stock-on-hand and unit costs. No PDF export implementation was found in the dashboard; this standalone report supplies the report deliverable.

Recommended workflow

- Begin with the executive overview and check each metric's evaluation window.
- Use segmentation and churn outputs to support analyst review, not automatic treatment without checks.
- Inspect demand-confidence flags before acting on inventory recommendations.
- Retain dataset windows and model versions with exported decisions.

11. Acceptance Criteria Status

Target	Current evidence	Status
Forecast MAPE ≤12%	Prophet MAPE 27.50% on non-zero days	Not met
Churn ROC-AUC ≥0.88	Optuna temporal ROC-AUC 0.7695	Not met
Precision@top 20% ≥0.75	Optuna temporal 0.7555; baseline temporal 0.7381	Validation-dependent
Inventory reduction 25–40%	No measured intervention outcomes	Not measured
Stockout reduction 30–50%	No stock/stockout labels	Not measured
Revenue improvement 15–25%	No controlled impact evaluation	Not measured
Configured data checks	36 pass; 0 fail	Completed
Public HTTPS demo	Deployment not verified	Outstanding
Docker/Kubernetes/CI/CD/Airflow	Not fully completed and validated	Outstanding

Business impact is marked not measured, rather than a failed experiment: required operational labels and controlled intervention results are unavailable.

12. Architecture and Methods

Layer	Implementation
Input	Excel workbook with two historical retail sheets.
Preparation	Python/Pandas cleaning, feature engineering and quality validation.
Analytics	EDA, RFM transformations, K-Means and DBSCAN.
Forecasting	Daily series, Prophet, PyTorch Lightning LSTM, holdout comparison and ensemble.
Churn	XGBoost, temporal snapshots, Optuna, SHAP and feature importance.

Layer	Implementation
Inventory	Historical product demand rates, review categories and confidence audit.
Monitoring	Evidently reports and local experiment/artifact records.
Presentation	Streamlit dashboard with analytical views and downloads.

Methodological notes

- Forecasting uses a chronological final 30-day holdout; random-split churn results are separated from temporal validation.
- MAPE excludes zero-actual days; WAPE is included as a complementary measure.
- K-Means k=6 was selected for interpretability despite a higher silhouette at k=2.
- No causal claims or unsupported claims of revenue improvement, savings, or production scale are made.

13. Risks, Limitations and Responsible Use

Customer identity gaps

Missing IDs may bias customer analysis toward identifiable customers.

Historical age

Records end in December 2011 and may not represent current behaviour.

Bulk-order influence

Extreme quantities can dominate demand rates and require review.

Model generalisation

Metrics rely on limited holdout windows; evaluate on newer data.

Operational inputs

Stock, costs, lead-time variability and stockout observations are unavailable.

Monitoring design

Some comparison windows have caveats and do not prove model failure.

Decision support

Predictions should support proportionate human review; interventions require evaluation.

14. Recommended Next Steps

Priority	Action	Reason
P0	Validate dashboard pages and capture demo screenshots.	Confirms user-facing behaviour.
P0	Complete README, setup instructions and reproducible commands.	Improves review and reproducibility.
P0	Review this PDF for page count, readability and assets.	Required report artifact.
P1	Deploy a public HTTPS demo and test paths/memory use.	Required demo deliverable; not verified.
P1	Record a 4–8 minute demo across the analytical pages.	Required submission artifact.
P1	Improve forecasting with additional validation before changing ensemble weights.	Forecast target is unmet.
P1	Obtain stock, cost, lead-time and stockout data.	Needed for operational impact measurement.
P2	Expand temporal churn validation across cutoffs and cohorts.	Reduces reliance on one test period.
P2	Finish and test retraining and deployment automation.	Closes production-readiness gaps.

15. Conclusion

RetailPulse demonstrates a multi-stage retail analytics workflow across more than one million historical transaction rows. Completed components include cleaning, configured quality validation, EDA, RFM segmentation with PCA and DBSCAN, forecast baselines, churn modelling with temporal evaluation and SHAP, historical-demand inventory review, monitoring diagnostics, and a multi-page dashboard.

Prophet is the best current forecast baseline, with 27.50% MAPE on non-zero holdout days and 37.02% WAPE. The Optuna-tuned temporal churn model achieved ROC-AUC 0.7695 and precision@top 20% of 0.7555 at the evaluated cutoff. Forecast and churn targets are not all met. Inventory impact remains unmeasured because operational data and intervention outcomes are missing.

The appropriate conclusion is an analytical prototype with transparent limitations, not a fully deployed production platform or a proven source of quantified business improvement.

Project artifacts

- Cleaned transactions and daily sales series in data/processedData.
- EDA, segmentation, forecasting, churn, inventory, validation and monitoring reports in reports/modelReports.
- Analytical PNG figures in reports/figures.
- Streamlit dashboard source in app.py.
- Local experiment records and model artifacts for reproducibility.

Generated 19 October 2026. Confirm the final PDF page count and visual quality before submission.